

Research Profile

I design and build **interactive AI systems for health and well-being**. My work centers on **human-AI co-sensemaking**: keeping people the authors of what they come to understand, rather than recipients of what a model concludes. Working across **multimodal sensing**, **interactive visual analytics**, and **generative AI**, I build systems that let individuals and domain experts question, revise, and own the interpretations they act on. My research lies at the intersection of **Ubiquitous Computing (UbiComp)**, **Human-Computer Interaction (HCI)**, and **Health Informatics**.

Keywords: Human-AI Co-Sensemaking, Multimodal Health Sensing, Generative AI Systems, Clinical Decision Support, Interactive Visual Analytics, Digital Well-being

Employment

Postdoctoral Research Scientist	Department of Biomedical Informatics, Columbia University Advisor: <u>Prof. Xuhai “Orson” Xu</u>	2026–Present
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Education

Ph.D.	The University of Hong Kong , Department of Electrical and Computer Engineering Advisor: Prof. Edith C.H. Ngai Thesis: “ <i>Empowering Pervasive Healthcare: Mobile Analytics Systems Leveraging Multimodal Data</i> ” (Excellent PhD Thesis)	2021–2025
M.E.	Xiamen University , Department of Computer Science and Technology Advisor: Prof. Longbiao Chen Thesis: “ <i>Traffic Violation Hotspot Identification, Diagnosing, and Prediction Based on Spatial Crowdsensing Big Data</i> ” (Outstanding Graduate Thesis)	2018–2021
B.E.	Xiamen University Major: Computer Science and Technology, School of Informatics Thesis: “ <i>Identification and Visualization of Driving Violations Based on Large-Scale Vehicle Trajectory Data</i> ” (Outstanding Undergraduate Thesis) Minor: Mathematical Finance, The Wang Yanan Institute for Studies in Economics	2014–2018

Publications (* indicates equal contribution, † indicates corresponding author)

Full Peer-Reviewed Conference & Journal Papers

2026

- P35. [ACM HEALTH’26] **Zhihan Jiang***, Running Zhao*, Lin Lin, Yue Yu, Handi Chen, Xincheng Zhang, Xuhai Xu, Yifang Wang, Xiaojuan Ma, Edith C.H. Ngai. DietGlance: Dietary Monitoring and Personalized Analysis at a Glance with Knowledge-Empowered AI Assistant. *ACM Transactions on Computing for Healthcare*, 2026.
- P34. [CHI’26] **Zhihan Jiang**, Qianhui Chen, Chu Zhang, Yanheng Li, Ray LC. Hear You in Silence: Designing for Active Listening in Human Interaction with Conversational Agents Using Context-Aware Pacing. *ACM CHI Conference on Human Factors in Computing Systems*, 2026. ( **Honorable Mention Award, 5%**)
- P33. [CHI’26] Mengyuan Wu, **Zhihan Jiang***, Yang Fan*, Richard Feng, Sahiti Dharmavaram, Mathew Polowitz, Shawn Fallon, Bashima Islam, Lizbeth Benson, Irene Tung, David Creswell, Xuhai Xu. MindfulAgents: Personalizing Mindfulness Meditation via an Expert-Aligned Multi-Agent System. *ACM CHI Conference on Human Factors in Computing Systems*, 2026. ( **Honorable Mention Award, 5%**)
- P32. [TKDD’26] Jing Deng, Handi Chen*, **Zhihan Jiang***, Raymond Chi-Wing Wong, Edith C.H. Ngai. A Survey on Causality with Federated Learning: Challenges, Techniques, and Applications. *Transactions on Knowledge Discovery from Data*, 2026.

- P31. [ToN'26] Handi Chen, Jing Deng, Xiuzhe Wu, **Zhihan Jiang**, Xincheng Zhang, Xianhao Chen, Edith C.H. Ngai, Jiangchuan Liu. LiFeChain: Lightweight Blockchain for Secure and Efficient Federated Lifelong Learning in IoT. *IEEE Transactions on Networking*, 2026.

2025

- P30. [UIST'25] Running Zhao*, **Zhihan Jiang***, Xincheng Zhang, Chirui Chang, Handi Chen, Weipeng Deng, Luyao Jin, Xiaojuan Qi, Xun Qian, Edith C.H. Ngai. NoteIt: A System Converting Instructional Videos to Interactable Notes Through Multimodal Video Understanding. *The ACM Symposium on User Interface Software and Technology*, 2025.
- P29. [AFM'25] Ran An, **Zhihan Jiang**, Qiyu Cao, Shuang Zhang, Edith C. H. Ngai, Tianshuo Zhao. Chemically Programmable Fano Resonances via Colloidal Nanocrystal-Ligand Chemistry for Ultra-Sensitive Ion Detection. *Advanced Functional Materials*, 2025.
- P28. [TNSE'25] Zhiyi Zhong, Lin Lin, **Zhihan Jiang**, Xin Yuan, Edith C.H. Ngai, James Lam, Ka-Wai Kwok. Connectivity Determination Algorithm for Complex Directed Networks. *IEEE Transactions on Network Science and Engineering*, 2025.
- P27. [INFOCOM'25] Xincheng Zhang, Running Zhao, **Zhihan Jiang**, Handi Chen, Yulong Ding, Edith C.H. Ngai, Shuanghua Yang. Continual Learning with Strategic Selection and Forgetting for Network Intrusion Detection. *The 2025 IEEE International Conference on Computer Communications*, 2025.
- P26. [TMC'25] Running Zhao, Jiangtao Yu, Tingle Li, **Zhihan Jiang**, Chenwei Zhang, Cunshu Wu, Hang Zhao, Edith C.H. Ngai. SPACE: Speaker Adaptation for Acoustic Eavesdropping using mmWave Radio Signals. *IEEE Transactions on Mobile Computing*, 2025.
- P25. [IEEE Network'25] Handi Chen, Weipeng Deng, Shuo Yang, Jinfeng Xu, **Zhihan Jiang**, Edith C.H. Ngai, Jiangchuan Liu, Xue Liu. Towards Edge General Intelligence via Large Language Models: Opportunities and Challenges. *IEEE Network*, 2025.

2024

- P24. [JAACAPOpen'24] **Zhihan Jiang***, Adrienne Y.L. Chan*, Dawn Lum, Kirstie HTY Wong, Janice CN Leung, Patrick Ip, David Coghill, Rosa S Wong, Edith C.H. Ngai, Ian C.K. Wong. Wearable Signals for Diagnosing Attention-Deficit/Hyperactivity Disorder in Adolescents: A Feasibility Study. *Journal of American Academy of Child & Adolescent Psychiatry Open*, 2024.
- P23. [JFranklin'24] Xincheng Zhang, **Zhihan Jiang**, Yulong Ding, Edith C.H. Ngai, Shuang-Hua Yang. Anomaly Detection using Isomorphic Analysis for False Data Injection Attacks in Industrial Control Systems. *Journal of the Franklin Institute*, 2024.
- P22. [INFOCOM'24] Xincheng Zhang, Running Zhao, **Zhihan Jiang**, Zhicong Sun, Yulong Ding, Edith C.H. Ngai and Shuanghua Yang. AOC-IDS: Autonomous Online Framework with Contrastive Learning for Intrusion Detection. *The 2024 IEEE International Conference on Computer Communications*, 2024.
- P21. [TMC'24] Handi Chen, Rui Zhou, Yun-Hin Chan, **Zhihan Jiang**, Xianhao Chen, Edith C.H. Ngai. LiteChain: A Lightweight Blockchain for Verifiable and Scalable Federated Learning in Massive Edge Networks. *IEEE Transactions on Mobile Computing*, 2024.
- P20. [ICLR'24] Yun-Hin Chan, Rui Zhou, Running Zhao, **Zhihan Jiang**, Edith C.H. Ngai. Internal Cross-layer Gradients for Extending Homogeneity to Heterogeneity in Federated Learning. *The 12th International Conference on Learning Representations*, 2024.
- P19. [UbiComp/IMWUT'24] Yufei Wang, Wenting Zeng, Changzhen Liu, Zhuohan Ye, Jiawei Sun, Junxiang Ji, **Zhihan Jiang**, Xianyi Yan, Yongyi Wu, Yigao Wang, Dingqi Yang, Leye Wang, Daqing Zhang, Cheng Wang, Longbiao Chen. CrowdBot: An Open-Environment Robot Management System for On-Campus Services. *Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2024.

2023

- P18. [IoTJ'23] **Zhihan Jiang**, Vera van Zoest, Weipeng Deng, Edith C.H. Ngai, Jiangchuan Liu. Leveraging Machine Learning for Disease Diagnoses based on Wearable Devices: A Survey. *IEEE Internet of Things Journal*, 2023.
- P17. [TVCG/VIS'23] **Zhihan Jiang**, Handi Chen, Rui Zhou, Jing Deng, Xincheng Zhang, Running Zhao, Cong Xie, Yifang Wang, Edith C.H. Ngai. HealthPrism: A Visual Analytics System for Exploring Children's Physical and Mental Health Profiles with Multimodal Data. *IEEE Transactions on Visualization and Computer Graphics*, IEEE VIS 2023.

- P16. **[UbiComp/IMWUT'23] Zhihan Jiang**, Lin Lin, Xinchun Zhang, Jianduo Luan, Running Zhao, Longbiao Chen, James Lam, Ka Man Yip, Hung Kwan So, Wilfred HS Wong, Patrick Ip, Edith C.H. Ngai. A Data-Driven Context-Aware Health Inference System for Children during School Closures. *Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2023.
- P15. **[TCNS'23] Lin Lin, Zhihan Jiang**, Hong Lin, Edith C.H. Ngai, James Lam. On Quotients of Stochastic Networks over Finite Fields. *IEEE Transactions on Control of Network Systems*, 2023.
- P14. **[UbiComp/IMWUT'23] Tieqi Shou, Zhuohan Ye, Yayao Hong, Zhiyuan Wang, Hang Zhu, Zhihan Jiang**, Dingqi Yang, Binbin Zhou, Cheng Wang, Longbiao Chen. CrowdQ: Predicting the Queue State of Hospital Emergency Department Using Crowdsensing Mobility Data-Driven Models. *Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2023.

2022

- P13. **[TITS'22] Zhihan Jiang**, Xin He, Chenhui Lu, Binbin Zhou, Xiaoliang Fan, Cheng Wang, Xiaojuan Ma, Edith C.H. Ngai, Longbiao Chen. Understanding Drivers' Visual and Comprehension Loads in Traffic Violation Hotspots Leveraging Crowd-Based Driving Simulation. *IEEE Transactions on Intelligent Transportation Systems*, 2022.
- P12. **[Healthcare'22] Zhihan Jiang***, Ka-Man Yip*, Xinchun Zhang, Jing Deng, Wilfred Wong, Hung-Kwan So, Edith C. H. Ngai. Identifying the High-Risk Population for COVID-19 Transmission in Hong Kong Leveraging Explainable Machine Learning. *Healthcare*, 2022.
- P11. **[CHI'22] Chuhan Shi, Zhihan Jiang**, Xiaojuan Ma, Qiong Luo. A Personalized Visual Aid for Selections of Appearance Building Products with Long-term Effects. *Proc. of the ACM CHI Conference on Human Factors in Computing Systems*, 2022.
- P10. **[UIC'22] Jiannan Gao, Yigao Wang, Zhihan Jiang**, Hang Zhu, Qiyue Zhong, Xiaoliang Fan, Longbiao Chen, Cheng Wang. iTA: Inferring Traffic Accident Hotspots with Vehicle Trajectories and Road Environment Data. *The 19th IEEE International Conference on Ubiquitous Intelligence and Computing*, 2022.
- P9. **[TMC'22] Hang Zhu, Tieqi Shou, Ruiying Guo, Zhihan Jiang**, Zeyu Wang, Zhiyuan Wang, Zhiyong Yu, Weijie Zhang, Cheng Wang, Longbiao Chen. RedPacketBike: A Graph-Based Demand Modeling and Crowd-Driven Station Rebalancing Framework for Bike Sharing Systems. *IEEE Transactions on Mobile Computing*, 2022.

2021

- P8. **[TMC'21] Zhihan Jiang**, Hang Zhu, Binbin Zhou, Chenhui Lu, Mingfei Sun, Xiaojuan Ma, Xiaoliang Fan, Cheng Wang, Longbiao Chen. CrowdPatrol: A Mobile Crowdsensing Framework for Traffic Violation Hotspot Patrolling. *IEEE Transactions on Mobile Computing*, 2021.
- P7. **[JNCA'21] Longbiao Chen, Zhihan Jiang**, Dingqi Yang, Thi-Mai-Trang Nguyen, Cheng Wang. Fog Radio Access Network Optimization for 5G Leveraging User Mobility and Traffic Data. *Journal of Network and Computer Applications*, 2021.
- P6. **[UbiComp/IMWUT'21] Longbiao Chen, Chenhui Lu, Fangxu Yuan, Zhihan Jiang**, Leye Wang, Daqing Zhang, Ruixiang Luo, Xiaoliang Fan, Cheng Wang. UVLens: Urban Village Boundary Identification and Population Estimation Leveraging Open Government Data. *Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, 2021.
- P5. **[UIC'21] Xin He, Hong Hong, Yanwen Liu, Xiaojuan Ma, Zhihan Jiang**, Longbiao Chen, Ming Cheng, Cheng Wang, Yongchuan Li. CovidPass: A Contactless Check-in System for Keeping Social Distance in Public Health Crisis. *The 18th IEEE International Conference on Ubiquitous Intelligence and Computing*, 2021.

Previous

- P4. **[ISJ'20] Zhihan Jiang**, Longbiao Chen, Binbin Zhou, Jingchun Huang, Tianqi Xie, Xiaoliang Fan, Cheng Wang. iTV: Inferring Traffic Violation-Prone Locations with Vehicle Trajectories and Road Environment Data. *IEEE Systems Journal*, 2020.
- P3. **[FCS'20] Zhihan Jiang**, Yan Liu, Xiaoliang Fan, Cheng Wang, Jonathan Li, Longbiao Chen. Understanding Urban Structures and Crowd Dynamics Leveraging Large-Scale Vehicle Mobility Data. *Frontiers of Computer Science*, 2020.
- P2. **[UIC'19] Ruiying Guo, Zhihan Jiang**, Jingchun Huang, Jianrong Tao, Cheng Wang, Jonathan Li, Longbiao Chen. BikeNet: Accurate Bike Demand Prediction Using Graph Neural Networks for Station Rebalancing. *The 16th IEEE International Conference on Ubiquitous Intelligence and Computing*, 2019.

- P1. [EWSN'19] Longbiao Chen, **Zhihan Jiang**, Jiangtao Wang, Yasha Wang. Data-Driven Bike Sharing System Optimization: State of the Art and Future Opportunities. *The 2019 International Conference on Embedded Wireless Systems and Networks*, 2019.

Peer-Reviewed Workshop, Poster & Short Papers

- W6. [UbiComp/ISWC Workshop'26] Rosemary Li, Nuo Chen, Shenghan Gao, **Zhihan Jiang**. A Triangulate-and-Audit Framework for Measuring Depression-Indicative Self-Disclosure in YouTube Comments. *WellComp: International Workshop on Computing for Well-Being @ ACM UbiComp/ISWC*, 2026.
- W5. [Education Forum] **Zhihan Jiang**, Xuhai “Orson” Xu. From AI-as-Shortcut to AI-as-Partner: Student-AI Collaborative Inquiry for Health-Agent Education. *UbiComp/ISWC 2026 Education Forum*.
- W4. [Tutorial] **Zhihan Jiang**, Will Ke Wang, Blue (Georgianna) Lin, Brenna Li, Xuhai “Orson” Xu. From Multimodal Sensing Data to Actionable Health Insights: A Hands-on Guide to Prototyping Your Personal LLM Health Agent. *ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC)*, 2026.
- W3. [CHI Workshop'26] **Zhihan Jiang**, Mengyuan “Millie” Wu, Ruishi Zou, Shiyu Xu, Emma Macmanus, Steven Liao, Ping Zhang, Dakuo Wang, James L. David, Nabila El-Bassel, Lena Mamykina, Frances R. Levin, Ryan Sultan, Xuhai Xu. PAWS: Empowering Everyday Cannabis Use Disorder Support through a Personalized AI Digital Pet on Smartwatches. *Everyday Wearable for Personalized Health and Well-Being Workshop @ ACM CHI Conference on Human Factors in Computing Systems*, 2026. (🏆 **Best Poster Award**)
- W2. [ICASSP Workshop'23] **Zhihan Jiang**, Cong Xie, Edith C.H. Ngai. A Health Profiling Framework for Children Leveraging Multimodal Learning Based on Ambient Sensor Signals. *Ambient AI Workshop @ IEEE International Conference on Acoustics, Speech, and Signal Processing*, 2023.
- W1. [HHME'18] Yiyang Lai, **Zhihan Jiang**, Xiaoliang Fan, Cheng Wang, Jun Li, Longbiao Chen. Post-Disaster Traffic Visualization and Analysis based on Large-Scale Vehicle Trajectory Data. *The 14th Joint Conference on Harmonious Human-Machine Environment*, 2018.
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Patents

- T3. Longbiao Chen, **Zhihan Jiang**, Xiaoliang Fan, Cheng Wang, Hong Hong. A Method, System, and Readable Storage Medium for Patrol Route Planning Based on Traffic Violation Hotspot Prediction. China Patent CN202010693026.4.
- T2. Longbiao Chen, **Zhihan Jiang**, Cheng Wang. A Method and System for Traffic Violation Identification Based on Traffic Data and Street View Data. China Patent CN202010031165.0.
- T1. Longbiao Chen, Ruiying Guo, **Zhihan Jiang**, Zhiyuan Wang, Cheng Wang. A Method for Bike Demand Prediction and Scheduling Based on Deep Learning and Crowd Sensing. China Patent CN202010599191.3.
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Academic Service

Editor, Chair, Committee

- **Lead Organizer**, Tutorial on Prototyping Personal LLM Health Agents @ ACM UbiComp/ISWC 2026
- **Technical Program Committee** @ EIoT Workshop, UbiComp/ISWC 2026
- **Social Chair / Technical Program Committee** @ WellComp Workshop, UbiComp/ISWC 2026
- **Session Chair** @ CHI 2026
- **Program Committee** @ UbiComp/ISWC 2026 Poster Track

Journal Reviewer

- [IMWUT] Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
- [HEALTH] ACM Transactions on Computing for Healthcare
- [JMIR] Journal of Medical Internet Research
- [TVCG] IEEE Transactions on Visualization and Computer Graphics
- [TAFFC] IEEE Transactions on Affective Computing
- [TMC] IEEE Transactions on Mobile Computing
- [TITS] IEEE Transactions on Intelligent Transportation Systems

- [TII] IEEE Transactions on Industrial Informatics
- [TCE] IEEE Transactions on Consumer Electronics
- [TCSS] IEEE Transactions on Computational Social Systems
- [JNCA] Journal of Network and Computer Applications
- [IoTJ] IEEE Internet of Things Journal
- [EAAI] Engineering Applications of Artificial Intelligence

Conference Reviewer

- [VIS] IEEE Visualization and Visual Analytics Conference (*reviews recognized as highly useful*) 2026
- [UIST] The ACM Symposium on User Interface Software and Technology 2026
- [IUI] ACM Conference on Intelligent User Interfaces 2026
- [PacificVis] IEEE Pacific Visualization Conference (*reviews recognized as highly useful*) 2026
- [HAI] International Conference on Human-Agent Interaction 2026
- [IJCNN] International Joint Conference on Neural Networks 2025
- [CHI] ACM Conference on Human Factors in Computing Systems 2025–2026
- [TEI] ACM Int. Conf. on Tangible, Embedded, and Embodied Interaction 2025
- [INFOCOM] IEEE International Conference on Computer Communications 2023–2025
- [SenSys] ACM Conference on Embedded Networked Sensor Systems 2022–2025
- [ICASSP] IEEE Int. Conf. on Acoustics, Speech, and Signal Processing 2024–2026
- [ICDCS] IEEE International Conference on Distributed Computing Systems 2022–2023
- [IWQoS] IEEE/ACM International Workshop on Quality of Service 2023
- [UIC] IEEE Int. Conf. on Ubiquitous Intelligence and Computing 2020–2022
- [ChineseCSCW] Chinese Conference on CSCW and Social Computing 2020
- [ICC] IEEE International Conference on Communications 2020

Selected Scholarships & Honors

Best Poster Award , Everyday Wearable for Personalized Health and Well-Being Workshop @ CHI 2026	2026
Honorable Mention Award (×2), CHI 2026	2026
OPA Academic Conference Travel Grant Award , Columbia University	2026
Google Cloud Research Credit Award	2025
Excellent PhD Thesis , The University of Hong Kong	2025
HKU Foundation First Year Excellent PhD Award (one of no more than 15 recipients university-wide)	2024
IEEE VIS Inclusivity and Diversity Scholarship , IEEE VIS	2023
Conference Grant for RPg Student (twice), The University of Hong Kong	2023
Outstanding Graduate Thesis , Xiamen University	2021
Outstanding Graduates , Xiamen University	2021
National Scholarship (highest honor awarded by the Ministry of Education of China, top 1%)	2020
Outstanding Undergraduate Thesis , Xiamen University	2018
Academic Innovation Scholarship , Xiamen University	2017
Academic Excellence Scholarship (four times), Xiamen University	2015–2018

Teaching Experience

Teaching Roles and Responsibilities

- Leading tutorials, designing and grading assignments, and holding office hours.
- Certificate in Teaching and Learning in Higher Education awarded by The University of Hong Kong.

Conference Tutorial

From Multimodal Sensing Data to Actionable Health Insights: A Hands-on Guide to Prototyping UbiComp/ISWC, Oct 2026
Your Personal LLM Health Agent (Lecturer)

Guest Lecturer, Columbia University

BINF 4070 / COMS 4995: The Future of Personal Health Assistant: Multimodal Health Data Fall 2026
Management Agents (graduate course); lecture on LLM agents, tool use, and function calling

Teaching Assistant, The University of Hong Kong

COMP3234 / ELEC3443: Computer and Communication Networks
Undergraduate core course, ~70–100 students per year Fall 2021–2024

Teaching Assistant, Xiamen University

Software Architecture and Design Patterns Spring 2019

Invited Talks & Seminars

- Research Summit for Egocentric Intelligence with Project Aria, Meta Reality Labs Research Redmond, WA, USA, Jul 2026
- Southern University of Science and Technology Shenzhen, China, Jun 2026
- Northeastern University Human-Centered AI Lab (NEU HAI) Boston, USA, May 2026
- Dept. of Computer and Computing Science, Hangzhou City University Hangzhou, China, Oct 2025
- Tam Wing Fan Innovation Wing Two, The University of Hong Kong Hong Kong SAR, Jun 2024
- Dept. of Electrical and Computer Engineering, The University of Hong Kong Hong Kong SAR, Oct 2023
- MARS Group, Xiamen University Xiamen, China, Aug 2023

Selected Media Coverage

- **Featured Interview: “Her Power in Science: Ubiquitous Computing Empowers Healthcare”** Mar 2024
Hong Kong Wen Wei Po (Mainstream Press) & The University of Hong Kong (Official Multi-channel Feature)
Selected as a featured doctoral candidate showcasing the societal value of pervasive healthcare research.
- **Research Highlight: AI-Powered Note-taking Tool “NoteIt”** Sep 2025
The University of Hong Kong (Official X/Twitter Feature)
Featured coverage of the team’s innovative multimodal video understanding system.
- **Research Highlight: AI System for Child Health Assessment** Dec 2023
The University of Hong Kong (Official X/Twitter Feature)
Highlighted contributions to designing health inference systems during COVID-19 school closures.